



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SENGAL ANNAI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

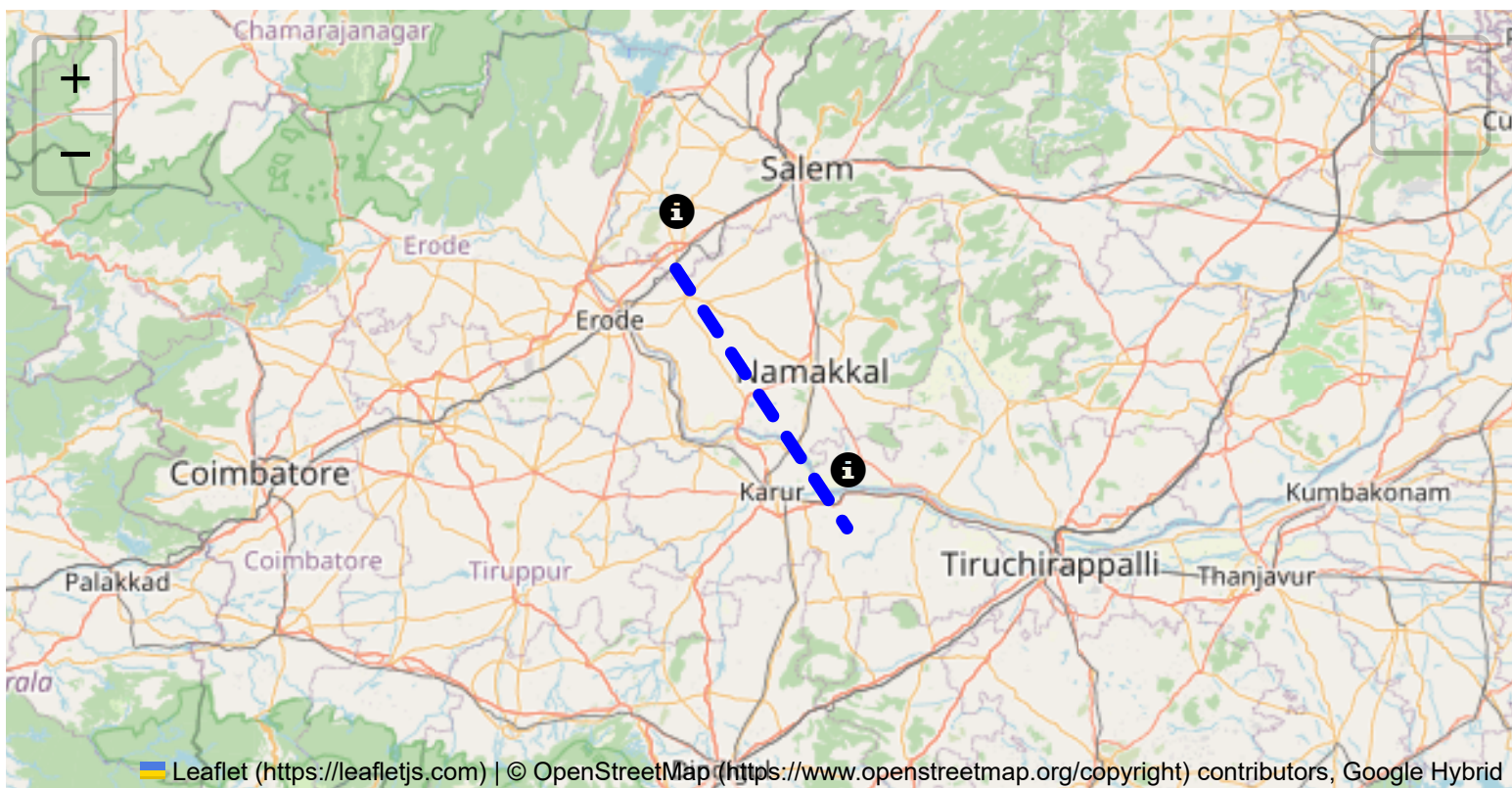
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

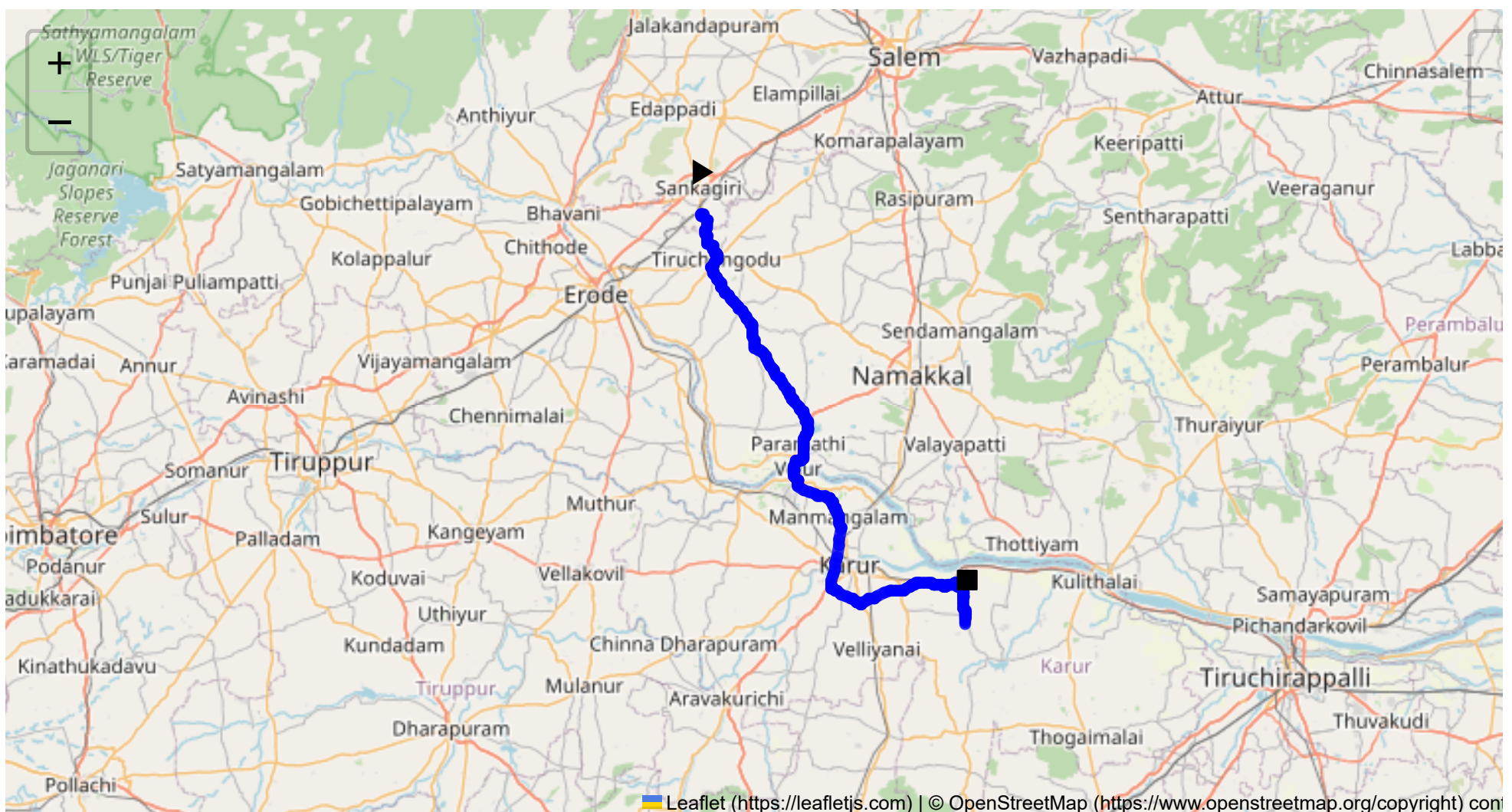
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 95.83 km
Estimated Duration: 1.8 hours
Adjusted Duration (Heavy Vehicle): 2.3 hours
Start: (11.4381, 77.8734)
End: (10.880237, 78.242837)



Welcome to the Journey Risk Management Study

Here's a comprehensive analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India to V6JV+33X, MDR632, Sengal, Tamil Nadu 639102, India:

1. Overview of the Route Map:

- The route primarily involves taking National Highways and State Highways. It covers a distance of approximately 95.83 kilometers, generally requiring around 1.83 hours for heavy vehicles. Key

highways may include NH44 and state roads leading to MDR632.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate. The region encounters monsoons typically from June to September, leading to potential heavy rainfall and waterlogging. This can create slippery road conditions and reduced visibility. Summer months can see extreme heat, affecting vehicle performance.

3. Traffic Patterns and Congestion-Prone Areas:

- The highways are busy, with peak traffic generally observed during early morning hours (7-9 AM) and late afternoon (4-6 PM). Possible congestion areas include urban settlements near Salem and Namakkal. Markets and local events might occasionally exacerbate traffic conditions.

4. Assessment of Road Quality and Infrastructure:

- National Highways in Tamil Nadu are generally well-maintained, but vigilance is needed in rural state highways where road conditions may vary. Watch out for narrow roads, sharp bends, and potholes, particularly after monsoon seasons.

5. Suggestions for Alternative Routes for Emergencies:

- Alternative routes might include using different National Highways or diverting through minor roads if traffic or incidents occur. Route planning tools with live traffic updates can better guide real-time diversions.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transportation of hazardous materials is regulated under strict guidelines, mandating proper labeling and route permissions. Specific timings and routes are sometimes restricted, favoring night transport to minimize public interaction.

7. Overview of Historical Incidents:

- There are documented accidents involving heavy vehicles near industrial regions en route, often due to poor visibility, overloading, or loss of control. Local news archives and road safety audits provide incident insights for such routes.

8. Environmental Considerations and Sensitive Areas:

- The route passes through agricultural and semi-urban settlements. Sensitivity to air and noise pollution is crucial, especially during transport of hazardous materials. Avoid spills and loud operations near these zones.

9. Analysis of Communication Coverage:

- Generally, network coverage along highways in Tamil Nadu is robust. However, coverage might be spotty in certain rural patches; it's advisable to carry multiple communication tools and ensure devices are well-charged.

10. Estimated Emergency Response Times for Different Route Segments:

- Larger towns like Salem have quicker emergency responses due to better infrastructure and proximity to hospitals and fire stations. Rural segments may experience delayed responses, up to 30-45 minutes, depending on accessibility.

11. Overall Summary of Risk Assessment:

- The route is manageable for heavy vehicles but requires caution concerning weather conditions, traffic congestion, and local regulations. Driver preparedness for emergency diversions and proactive communication are necessary. Proper vehicle maintenance and adherence to safety guidelines for hazardous materials are crucial to minimizing risk.

This analysis indicates a moderate risk level, necessitating careful planning and vigilance during transit. Regular updates and virtualization tools could greatly aid in ensuring a safer trip for drivers and their cargo.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16755, 78.01568	30 KM/Hr
9	Turn	Medium	11.15579, 78.02169	30 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14485, 78.02282	15 KM/Hr
12	Turn	High	10.93639, 78.22893	15 KM/Hr
13	Turn	Medium	10.93596, 78.22951	30 KM/Hr
14	Turn	High	10.93157, 78.23134	15 KM/Hr
15	Turn	Medium	10.92831, 78.23741	30 KM/Hr
16	Turn	High	10.91770, 78.23846	15 KM/Hr
17	Turn	Medium	10.91716, 78.23893	30 KM/Hr
18	Turn	Medium	10.90095, 78.23910	30 KM/Hr
19	Turn	High	10.89479, 78.24318	15 KM/Hr
20	Turn	Medium	10.89058, 78.24321	30 KM/Hr
21	Turn	Medium	10.88993, 78.24283	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
21	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.92958, 78.05452



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16755, 78.01568



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15579, 78.02169



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14485, 78.02282



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.93639, 78.22893



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.93596, 78.22951



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.93157, 78.23134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.92831, 78.23741



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.91770, 78.23846



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.91716, 78.23893



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.90095, 78.23910



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.89479, 78.24318



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89058, 78.24321



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.88993, 78.24283
